

Jalal Abdel Halim

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Summary

PhD Candidate in Computer Science specializing in Cyber Security and Machine Learning.
Proficient in setting up and managing Proxmox servers and deploying containerized solutions.
Skilled in Linux command-line operations, system administration, and automation.

Education

Doctor of Philosophy - PhD . Computer Science, 2023 - May 2028 (Expected)

University of Toledo, Toledo, OH

GPA: 3.8/4.0

Master of Science in Cyber Security, 2021 - May 2023

University of Toledo, Toledo, OH

GPA: 3.7/4.0

Bachelor of Science in Information Technology, 2017-2021

University of Toledo, Toledo, OH

GPA: 3.5/4.0

Experience

Research/Teaching Assistant, The University of Toledo, Toledo, OH; Aug 2021 – Present

- Provided hands-on troubleshooting and support for students, evaluating assignments and communicating complex concepts clearly.
- Streamlined backend lab infrastructure, reducing environment setup time from several hours to under 30 minutes to improve operational efficiency.
- Managed concurrent environments and access for 55+ active users across two courses, ensuring secure and reliable daily operations.
- Developed a real-time detection tool utilizing data analysis to identify issues and protect end-users from threats.
- Administered daily system configurations and handled routine hardware and network troubleshooting to maintain reliable service.

Information Technology Intern, O-I, Perrysburg, OH; May 2022 – Aug 2022

- Utilized specialized software platforms to effectively manage access points and maintain secure device operations.
- Analyzed data from multiple internal systems (Rapid7, ServiceNow) to create detailed reports utilizing PowerBI.
- Developed automated scripts to extract data, significantly improving administrative task efficiency and overall accuracy.
- Conducted meetings and collaborated directly with global team members to coordinate personal and group projects.

Publications

U. B. Guntupalli, M. Pujari, J. Abdel Halim, W. Sun, "Investigating the Influence of Black-Box Adversarial Machine Learning Attacks in Live Network Environments," Proceedings of the 12th International Conference on Information Systems Security and Privacy (ICISSP), SciTePress, pp. 462-473, 2026.

- Implemented a live network validation pipeline to evaluate the robustness of machine learning-based intrusion detection systems against black-box adversarial attacks, providing empirical evidence of practical exploitability.

J. Abdel Halim, W. Sun, "A Cost-Effective Tool for Senior Activity," *AI Revolution: Research, Ethics and Society (AIR-RES 2025)*, Springer Nature, p. 267, Jan. 2026.

- Developed a privacy-preserving tracking system featuring a custom web interface and 3D data visualization to remotely monitor daily activities.

J. Abdel Halim, W. Sun, "Empowering users with an effective tool for social media spam detection," *Cybersecurity: Cyber Defense, Privacy and Cyber Warfare*, De Gruyter Brill, pp. 333-354, Nov. 2025.

- Engineered an end-to-end machine learning pipeline capable of multimodal (text, image, and video) spam detection across various platforms.

J. Abdel Halim, W. Sun, "Towards building a versatile tool for social media spam detection," *2023 Congress in Computer Science, Computer Engineering, & Applied Computing (CSCE)*, IEEE, pp. 2378-2384, Jul. 2023.

- Designed and trained machine learning classifiers to process social media text and images in real-time, providing users with proactive security alerts.

Certifications

Google IT Support Specialization - Coursera (Issued July 2021)

ID: PM5MKQWLFKTT

EXIN DevOps Foundation – EXIN (Issued May 2021)

ID: 6390916.20814718

ITIL Foundation Certificate in IT Service Management - AXELOS Global Best Practice (Issued May 2021)

ID: GR671270843JA

Skills

- **Infrastructure:** Proxmox, Containerization, Kali Linux, Ubuntu Server, ESXi.

- **Security:** Penetration Testing, Fortinet, FortiNAC, Rapid7.
- **Languages/Tools:** Python, Java, Machine Learning, PowerBI, Selenium.

Honors & Awards

College of Engineering Dean's list - Associate Dean of Undergraduate Studies

May 2021